

An intro to the bioinformatic web

https://bioinfmsc8.bio.ed.ac.uk/AY25_BPSM07.html

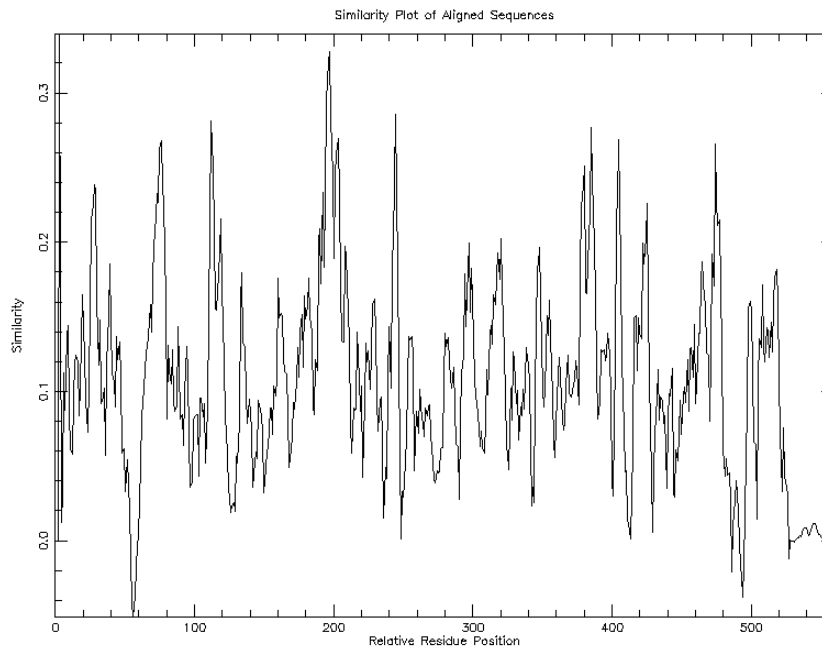
Quick links:
Click on any ↗ to get back here!

- it's a conservation plot!
- different viewpoints
- goals for today
- NCBI
- NCBI taxonomy: Nematoda
- doing a more specific search
- doing a more specific search NOT partial
- a single sequence record
- downloading the sequences
- EBI Clustal Omega for aligning the sequences
- **EMBOSS** for many sequence tools
- **plotcon** on our server
- **clustalo** on our server
- NCBI's eutils 1: esearch
- NCBI's eutils 2: efetch
- Installing edirect
- git time again!

Exercises

1. Install edirect
2. Repeat the lecture process
3. Repeat the analysis for a different taxonomic group
4. Flowchart the process and work out how to automate it

📌What is this!?



It's a map of amino acid conservation built from an alignment of **cytochrome oxidase 1 (COX1)** gene sequences from many nematode worms.

"In evolutionary biology, conserved sequences are identical or similar sequences in nucleic acids (DNA and RNA) or proteins across species (orthologous sequences), or within a genome (paralogous sequences), or between donor and receptor taxa (xenologous sequences). Conservation indicates that a sequence has been maintained by natural selection.

A highly conserved sequence is one that has remained relatively unchanged far back up the phylogenetic tree, and hence far back in geological time. Examples of highly conserved sequences include the RNA components of ribosomes present in all domains of life, the homeobox sequences widespread amongst Eukaryotes, and the tmRNA in Bacteria. The study of sequence conservation overlaps with the fields of genomics, proteomics, evolutionary biology, phylogenetics, bioinformatics and mathematics."

🏠 How can we re-generate the conservation plot? Different viewpoints...

Biologist

How can I download some DNA sequences?

For a particular taxonomic group?

For a particular gene?

How can I align them?

How can I find regions of high/low conservation?

How can I figure out what these regions might do?

Bioinformatician

How can I automate this for any gene?

For any taxon?

How can I automatically repeat the analysis when new sequences are released?

How can I make it faster? more reliable?

Can I write code to enable other people to run this analysis on their data?

📌Goals for today

1. (re-)introduce you to some key tools and resources: trying different ways to get sequences from NCBI:
 - web browser: point and click in a graphical interface
 - web-like commands in a terminal
 - dedicated programmes
2. decide how to put those resources together to answer a biological question
3. try out our proposed solution and see if it works
4. more thinking about pipelines/workflows: how's things coming along with the optional [MyFirstPipeline](#) ICA ?! Need to be able to do that before we move onto the Python half of the course!!
5. more use of git and GitHub

🏠 **NCBI: National Centre for Biotechnology Information**
<https://www.ncbi.nlm.nih.gov/>

Which sections might appeal (more) to a biologist?
Which sections might appeal (more) to a bioinformatician?

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 **National Library of Medicine**
National Center for Biotechnology Information

Log in

All Databases ▼

Search

NCBI Home

Resource List (A-Z)

All Resources

Chemicals & Bioassays

Data & Software

DNA & RNA

Domains & Structures

Genes & Expression

Genetics & Medicine

Genomes & Maps

Homology

Literature

Proteins

Sequence Analysis

Taxonomy

Training & Tutorials

Variation

Welcome to NCBI

The National Center for Biotechnology Information advances science and health by providing access to biomedical and genomic information.

[About the NCBI](#) | [Mission](#) | [Organization](#) | [NCBI News & Blog](#)

Submit

Deposit data or manuscripts into NCBI databases



Download

Transfer NCBI data to your computer



Learn

Find help documents, attend a class or watch a tutorial



Develop

Use NCBI APIs and code libraries to build applications



Analyze

Identify an NCBI tool for your data analysis task



Research

Explore NCBI research and collaborative projects



Popular Resources

[PubMed](#)

[Bookshelf](#)

[PubMed Central](#)

[BLAST](#)

[Nucleotide](#)

[Genome](#)

[SNP](#)

[Gene](#)

[Protein](#)

[PubChem](#)

NCBI News & Blog

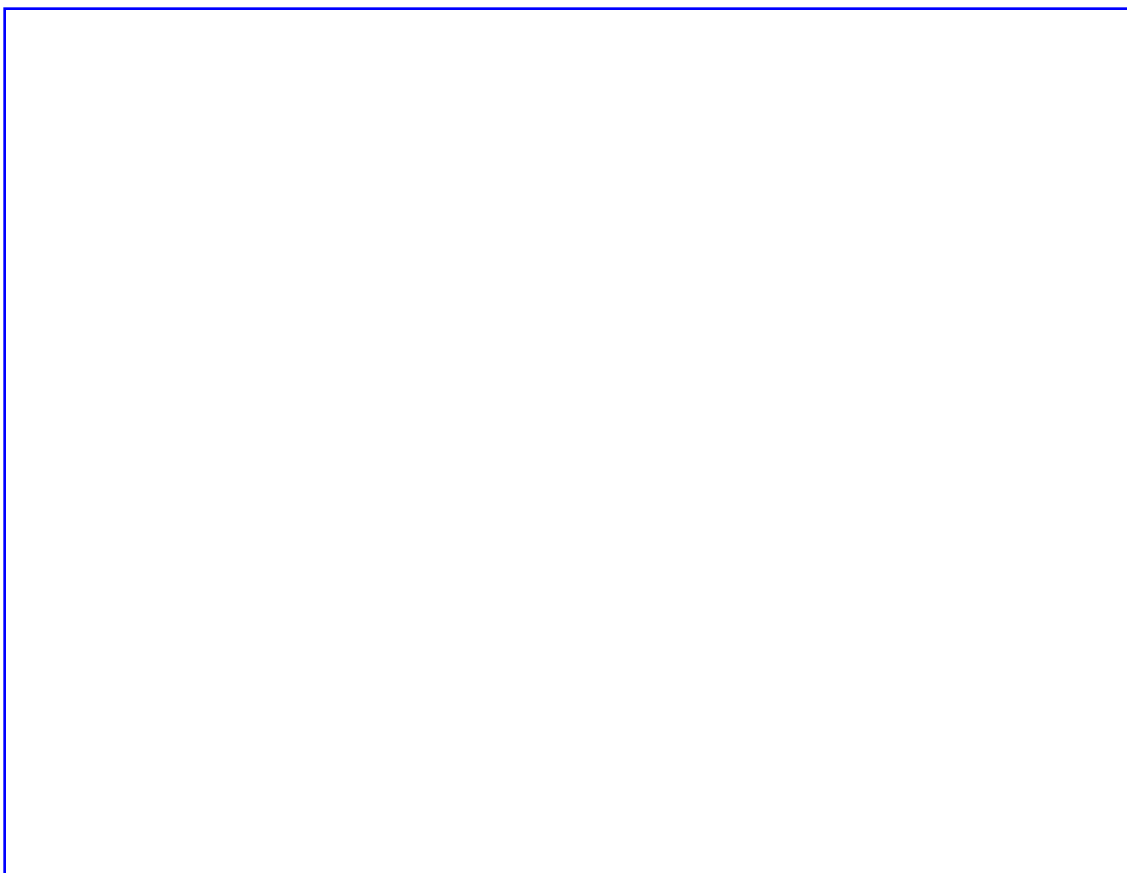
[NCBI Taxonomy Updates to Prokaryotes](#)
02 Oct 2024
[As previously announced, NCBI](#)

[Viewing Ligand-Protein Interactions in iCn3D](#)
30 Sep 2024
[Are you interested in viewing](#)

[Updated Genomes Terminology! "Representative Genome" is](#)

🏠 **NCBI taxonomy**

<https://www.ncbi.nlm.nih.gov/taxonomy>



🏠 Search for a taxon name: Nematoda

<https://www.ncbi.nlm.nih.gov/taxonomy/?term=nematoda>



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Taxonomy ▼

Display Settings: [Summary](#)

[Nematoda](#)

(nematodes), phylum, nematodes

Expand the level to view the hierarchy

<https://www.ncbi.nlm.nih.gov/Taxonomy/Browser/wwwtax.cgi?id=6231>

Taxonomy
Browser

[Try the New NCBI Taxonomy Pages!](#)
Explore our redesigned taxonomy browser and taxonomy record pages with faster, more intuitive search, taxonomy images, and links to NCBI Datasets and other data available at NCBI.

Entrez PubMed Nucleotide Protein Genome Structure PMC Taxonomy BioCollections

Search for as ☒ lock

levels using filter:

☐ BioProject
☐ db Var
☐ Identical Protein Groups
☐ Protein

☐ BioSample
☐ GEO DataSets
☐ Nucleotide
☐ SRA

☐ Conserved Domains
☐ Gene
☐ PubChem BioAssay
☐ Structure

☐ ClinVar
☐ GTR
☐ PMC
☐ Taxonomy

Lineage (full): [cellular organisms](#); [Eukaryota](#); [Opisthokonta](#); [Metazoa](#); [Eumetazoa](#); [Bilateria](#); [Protostomia](#); [Ecdysozoa](#)

- [Nematoda](#) (nematodes)
 - [Chromadorea](#)
 - [Enoplea](#)
 - [Nematoda incertae sedis](#)
 - [environmental samples](#)
 - [unclassified Nematoda](#)

Disclaimer: The NCBI taxonomy database is not an authoritative source for nomenclature or classification - please consult the relevant scientific literature for the most reliable information.

Reference: How to cite this resource - *Schoch CL, et al. NCBI Taxonomy: a comprehensive update on curation, resources and tools. Database (Oxford). 2020: baaa062. PubMed: 32761142 PMC: PMC7408187.*

Comments and questions to info@ncbi.nlm.nih.gov

Show numbers of database links

I chose to view nucleotide and protein sequence tallies: click the boxes, then click "Display"

- [Nematoda](#) (nematodes) [5,643,642](#) [2,632,476](#)
 - [Chromadorea](#) [5,420,236](#) [2,228,317](#)
 - [Araeolaimida](#) [895](#) [88](#)
 - [Axonolaimoidea](#) [894](#) [87](#)
 - [unclassified Araeolaimida](#) [2](#) [1](#)
 - [Chromadorida](#) [1,122](#) [165](#)
 - [Chromadoroidea](#) [1,105](#) [165](#)
 - [Paramicrolaimidae](#) [12](#)
 - [environmental samples](#) [1](#)
 - [unclassified Chromadorida](#) [5](#)
 - [Desmodorida](#) [1,168](#) [218](#)
 - [Desmodorina](#) [1,167](#) [218](#)
 - [environmental samples](#) [1](#)
 - [Desmoscolecida](#) [225](#) [2](#)
 - [Desmoscolecoidae](#) [225](#) [2](#)
 - [Monhysterida](#) [2,007](#) [211](#)
 - [Linhomoeina](#) [400](#) [23](#)
 - [Monhysterida incertae sedis](#) [6](#)
 - [Monhysterina](#) [1,599](#) [188](#)
 - [unclassified Monhysterida](#) [4](#)
 - [Plectida](#) [3,799](#) [515](#)
 - [Aegialoalaimidae](#) [6](#)
 - [Aulolaimidae](#) [2](#)
 - [Ceramomonematina](#) [40](#)
 - [Haliplectidae](#) [64](#) [13](#)
 - [Plectina](#) [3,689](#) [502](#)
-

Get overview for one taxon ID

<https://www.ncbi.nlm.nih.gov/Taxonomy/Browser/wwwtax.cgi?mode=Info&id=6231&lvl=3&p=nuccore&p=protein&lin=f&keep=1&srchmode=1&unlock>





Taxonomy
Browser

[Try the New NCBI Taxonomy Pages!](#)

Explore our redesigned taxonomy browser and taxonomy record pages with faster, more intuitive search, taxonomy images, and links to NCBI Datasets and other data available at NCBI.

EntrezPubMedNucleotideProteinGenomeStructurePMCTaxonomyBioCollections

Search for as complete name ☒ lock

Nematoda

Taxonomy ID: 6231 (for references in articles please use NCBI:txid6231)

current name

Nematoda

Genbank common name: **nematodes**

NCBI BLAST name: **nematodes**

Rank: **phylum**

Genetic code: [Translation table 1 \(Standard\)](#)

Mitochondrial genetic code: [Translation table 5 \(Invertebrate Mitochondrial\)](#)

Other names:

heterotypic synonym

Nemata

blast name

nematodes

common name(s)

Entrez records

Database name	Direct links	Subtree links	Links from type
BioProject	19	4,627	-
BioSample	654	93,204	-
Conserved Domains	17	170	-
GEO DataSets	-	45,723	-
Gene	-	213,639	-
Identical Protein Groups	-	2,259,778	-
Nucleotide	2	5,643,642	26
PubChem BioAssay	1	2,122	-
PMC	4,561	132,622	-
Protein	-	2,632,476	-
SRA	654	95,390	-
Structure	-	682	-

GenBank protein database

The query string limits the results to nematodes, and shows genera within the *Nematoda*, which we know already is taxonID 6231.

[https://www.ncbi.nlm.nih.gov/protein/?term=txid6231\[Organism:exp\]](https://www.ncbi.nlm.nih.gov/protein/?term=txid6231[Organism:exp])

How many proteins are there today?!

Over 2.6 million...

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Protein

Summary 20 per page Sort by Default order

TAXONOMY

Was this helpful? ☐ ☐

Thinking like a ...

Biologist

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How can I make it faster? more reliable?

Can I write code to enable other people to run this analysis on their data?

📌 We can add terms to make the query more specific

"OR" "AND" and "NOT" all do what you might expect...

txid6231[Organism:exp] AND (COX1[Gene Name] OR CO1[Gene Name])
is **NOT THE SAME AS**
txid6231[Organism:exp] AND (COX1 OR CO1)

[https://www.ncbi.nlm.nih.gov/protein/?term=txid6231%5BOrganism%3Aexp%5D+AND+\(COX1%5BGene+Name%5D+OR+CO1%5BGene+Name%5D\)](https://www.ncbi.nlm.nih.gov/protein/?term=txid6231%5BOrganism%3Aexp%5D+AND+(COX1%5BGene+Name%5D+OR+CO1%5BGene+Name%5D))

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Protein

Summary 20 per page Sort by Default order

See the [results of this search \(3 items\)](#) in our new [Identical Protein Groups](#) database.

Items: 1 to 20 of 22777

☐ [cytochrome c oxidase subunit 1, partial \(mitochondrion\)](#) [*Wiesoncholaimus jambio*]

See what it says the Search Details box on the right hand side of the web page!

TaxonID AND COX1 in Gene Name

<https://www.ncbi.nlm.nih.gov/protein/?term=txid6231%5BOrganism%3Aexp%5D+AND+COX1%5BGene+Name%5D>



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Protein

txid6231[Organism:exp] AND COX1[Gene Name]

Summary

20 per page

Sort by Default order

See the [results of this search \(3 items\)](#) in our new [Identical Protein Groups](#) database.

Items: 1 to 20 of 20655

☐

[cytochrome c oxidase subunit I, partial \(mitochondrion\), \[Peltamigratus christiei\]](#)

🏠 TaxonID AND COX1 in Gene Name, and NOT PARTIAL

<https://www.ncbi.nlm.nih.gov/protein/?term=txid6231%5BOrganism%3Aexp%5D+AND+COX1%5BGene+Name%5D+NOT+PARTIAL>



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Protein

Summary 20 per page Sort by Default order

See the [results of this search \(3 items\)](#) in our new [Identical Protein Groups](#) database.

Items: 1 to 20 of 827

☐ [cytochrome c oxidase subunit I \(mitochondrion\)](#) [Travassosius rufus]

TaxonID AND (COX1 OR CO1 in Gene Name), and NOT PARTIAL

[https://www.ncbi.nlm.nih.gov/protein/?term=txid6231%5BOrganism%3Aexp%5D+AND+\(COX1%5BGene+Name%5D+OR+CO1%5BGene+Name%5D\)+NOT+PARTIAL](https://www.ncbi.nlm.nih.gov/protein/?term=txid6231%5BOrganism%3Aexp%5D+AND+(COX1%5BGene+Name%5D+OR+CO1%5BGene+Name%5D)+NOT+PARTIAL)

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Protein

Summary 20 per page Sort by Default order

See the [results of this search \(3 items\)](#) in our new [Identical Protein Groups](#) database.

Items: 1 to 20 of 827

☐ [cytochrome c oxidase subunit I \(mitochondrion\)](#) [[Pristionchus pacificus](#)]

Thinking like a ...

Biologist

How can I download some DNA sequences?

~~For a particular taxonomic group?~~

~~For a particular gene?~~

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How can I find regions of high/low conservation?

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Bioinformatician

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For any taxon?

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How can I make it faster? more reliable?

Can I write code to enable other people to run this analysis on their data?

🏠 Viewing a single protein sequence record in NCBI <https://www.ncbi.nlm.nih.gov/protein/ADZ52296.1>

Contains

- "metadata"
- annotation of "things" in the sequence
- the sequence itself

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Protein

[GenPept](#)

cytochrome c oxidase subunit I (mitochondrion) [Pristionchus pacificus]

GenBank: ADZ52296.1
[Identical Proteins](#) [FASTA](#) [Graphics](#)

[Go to:](#)

LOCUS	ADZ52296	525 aa	linear	INV 25-JUL-2016
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We can view the information in several different formats

- GenPept: metadata and annotation and sequence. A complex, but strictly defined, structure (this is the default view we have just seen)
- Fasta (html): a brief description/header (the "product" and organism details) and the sequence. Page still has links, etc..

[https://www.ncbi.nlm.nih.gov/protein/ADZ52296.1?
report=fasta&log\\$=seqview](https://www.ncbi.nlm.nih.gov/protein/ADZ52296.1?report=fasta&log$=seqview)



- Fasta (text): a brief description in the header line (the "product" and organism details) and the sequence... nothing else...

[https://www.ncbi.nlm.nih.gov/protein/ADZ52296.1?
report=fasta&log\\$=seqview](https://www.ncbi.nlm.nih.gov/protein/ADZ52296.1?report=fasta&log$=seqview) **&format=text**

```
>ADZ52296.1 cytochrome c oxidase subunit I (mitochondrion) [Pristionchus pacificus]
MMNFKKYQGGLAVWLESSNHKDIGTLYFIFGLWGMVGTSLSLIIRLELAKPGFFLNGQLYNSVLTAAH
ILMIFFMVMPMSMIGGFGNWLLPLMLGAPDMSFRLNLSFWLLPTAMFLILDSSFVDMGCGTSWTVPPL
STMGHPGSSVDLAIFSLHCAGLSSILGGINFMCCTKNLRSSSISLEHMSLFVWTVFVTVFLVLVSLPVL
GAIITLLTDRLNTSFFDPSTGGNPLIYQHLFWFFGHPEVYILIPAFGIISQSTLYLTGKKEVFGSLGM
VYAILSIGLIGCVVWAHHMYTVGMDLDSRAYFTAATMVIAPVTGVKVFWSLATLFGMKMVFQPVLLWVLG
FIFLFTVGGLTGVLSSSLDIILHDTYVYVSHFHYVLSLGAVFGIPTGISLWNSFITGFVYDKMMSV
FILMFIGNLTFPPLHFAGLHGFPRKYVDYDPVYSIWNIVSSYSGMISTFALFLFIYVLMESFFSYRLVL
NDNFVNSSPEYSMSGYVFGHSYQSEIYFSSSLKH
```


**We need to download some sequences to do an alignment (to each other)....
The search we have decided on:**

txid6231[Organism:exp] AND (COX1[Gene Name] OR CO1[Gene Name]) NOT Partial

[https://www.ncbi.nlm.nih.gov/protein/?term=txid6231%5BOrganism%3Aexp%5D+AND+\(+COX1%5BGene+Name%5D+OR+CO1%5BGene+Name%5D\)+NOT+Partial](https://www.ncbi.nlm.nih.gov/protein/?term=txid6231%5BOrganism%3Aexp%5D+AND+(+COX1%5BGene+Name%5D+OR+CO1%5BGene+Name%5D)+NOT+Partial)

Just select the "Send to:" button to view the available choices

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Many data formats to choose from:



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Send to/Destination: File/Format:

We could use the accession list...(seems short, why?!)

< > ↻
🔖 ncbi.nlm.nih.gov/protein

YP_004300493.1
ADZ52296.1
XVF97250.1
XUY42425.1
XUY42413.1
XUS48701.1
XUB15251.1
XRT72422.1
XRT72410.1
XRT72398.1
XRT72386.1
XRT72374.1
XRT72362.1
XRT72350.1
XRT68368.1
XR058433.1
XQU46633.1
XQM64411.1
XQM41230.1
XQM41218.1

...to run lots of `wget` commands to get each sequence individually

```
wget -O YP_004300493.fasta "https://eutils.ncbi.nlm.nih.gov/entrez/eutils/efetch.fcgi?db=protein&rettype=fasta&retmode=text&id=YP_004300493.1"
wget -O ADZ52296.fasta "https://eutils.ncbi.nlm.nih.gov/entrez/eutils/efetch.fcgi?db=protein&rettype=fasta&retmode=text&id=ADZ52296.1"
....
```

...or perhaps have a simple script that uses that plain text file containing a list of accessions...

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Send to/Destination: File/Format: Accessions

...meantime, put the file in `${HOME}/Exercises/Lecture07/myfileofaccessions.txt`

Check the downloaded file contents

```
cd ${HOME}/Exercises/Lecture07/
```

```
head myfileofaccessions.txt
```

```
YP_004300493.1
ADZ52296.1
XVF97250.1
XUY42425.1
XUY42413.1
XUS48701.1
XUB15251.1
XRT72422.1
XRT72410.1
XRT72398.1
```

A simple bash script to get each one using `wget`

```
while read accession
do
echo -e "Downloading ${accession} ..."
wget -O ${accession}.fasta \
"https://eutils.ncbi.nlm.nih.gov/entrez/eutils/efetch.fcgi?db=protein&id=${accession}&strand=1&re
done < myfileofaccessions.txt
```

... and here is the output....

```
...
...
Downloading QDF64273.1 ...
--2025-10-02 15:19:00-- https://eutils.ncbi.nlm.nih.gov/entrez/eutils/efetch.fcgi?db=prot
Resolving eutils.ncbi.nlm.nih.gov (eutils.ncbi.nlm.nih.gov)... 130.14.29.110, 2607:f220:41
Connecting to eutils.ncbi.nlm.nih.gov (eutils.ncbi.nlm.nih.gov)|130.14.29.110|:443... conn
HTTP request sent, awaiting response... 200 OK
Length: unspecified [text/plain]
Saving to: 'QDF64273.1.fasta'

QDF64273.1.fasta          [ <=>          ]    643  --.-KB/s    in 0s
...
...
```



What do we have?

ls

ADZ52296.1.fasta	NP_008370.1.fasta	QRN72365.1.fasta	UXP34565.1.fasta	XRT72350.1.
AFV32091.1.fasta	NP_077265.1.fasta	QRN72378.1.fasta	UXP34577.1.fasta	XRT72362.1.
AFV32103.1.fasta	NP_579952.2.fasta	QRN72391.1.fasta	UXP34589.1.fasta	XRT72374.1.
AGQ46173.1.fasta	NP_579964.2.fasta	QRN72404.1.fasta	UYF12878.1.fasta	XRT72386.1.
AGV02082.1.fasta	NP_694905.1.fasta	QRN72417.1.fasta	UZH44202.1.fasta	XRT72398.1.
AOV93978.1.fasta	NP_851331.1.fasta	QRN72430.1.fasta	UZM11409.1.fasta	XRT72410.1.
AST14923.1.fasta	NP_954717.1.fasta	QRN72443.1.fasta	UZP15731.1.fasta	XRT72422.1.
AVV66502.1.fasta	QAA11073.1.fasta	QRN72456.1.fasta	WAP90821.1.fasta	XUB15251.1.
AXO78733.1.fasta	QAA11096.1.fasta	QRN72469.1.fasta	WAP91006.1.fasta	XUS48701.1.
AXS77374.1.fasta	QBK32731.1.fasta	QRN72482.1.fasta	WAR51552.1.fasta	XUY42413.1.
AYG51195.1.fasta	QCB91138.2.fasta	QRN72495.1.fasta	WAS08340.1.fasta	XUY42425.1.
AYU56870.1.fasta	QCB91139.1.fasta	QRN72508.1.fasta	WAS08352.1.fasta	XVF97250.1.
AYV63097.1.fasta	QCG69216.1.fasta	QRZ60650.1.fasta	WAS08364.1.fasta	YP_00102366
AZH08562.1.fasta	QCI56224.1.fasta	QST20096.1.fasta	WAS08413.1.fasta	YP_00155223
AZH08569.1.fasta	QCL17191.1.fasta	QTJ30044.1.fasta	WAX01707.1.fasta	YP_00170036
AZH08575.1.fasta	QCP88374.1.fasta	QWC93307.1.fasta	WAX01718.1.fasta	YP_00179539
AZU95939.1.fasta	QDE52612.1.fasta	QWK40675.1.fasta	WAX39110.1.fasta	YP_00190589
BAV78616.1.fasta	QDF64273.1.fasta	QXE43448.1.fasta	WBV76988.1.fasta	YP_00193646
BAV78636.1.fasta	QDF64304.1.fasta	QXJ80282.1.fasta	WCQ82417.1.fasta	YP_00272569
BAV78646.1.fasta	QEO33934.1.fasta	QXJ80295.1.fasta	WCQ82783.1.fasta	YP_00272571
BAV78658.1.fasta	QIE12490.1.fasta	QXJ80308.1.fasta	WCR50913.1.fasta	YP_00309710

BAV78670.1.fasta	QJQ35661.1.fasta	QXJ80321.1.fasta	WEB60571.1.fasta	YP_00309713
...				

Is that what we wanted? Probably not!

Clean up a bit....??

wc -l myfileofaccessions.txt

827 myfileofaccessions.txt

ls *.fasta | wc -l

827

rm *.fasta

ls *.fasta | wc -l

0

🏠...or (easiest for now) just use the NCBI file download facility to do it for us, into one file containing all the sequences we want in fasta format

 An official website of the United States government
[Here's how you know.](#)

Log in

Notice

Because of a lapse in government funding, the information on this website may not be up to date, transactions submitted via the website may not be processed, and the agency may not be able to respond to inquiries until appropriations are enacted. The NIH Clinical Center (the research hospital of NIH) is open. For more details about its operating status, please visit cc.nih.gov. Updates regarding government operating status and resumption of normal operations can be found at opm.gov.

Download, save the file as **cytoxseqs.fasta** (or whatever you want):
it's all plain text!

Assuming you have now copied the file to the server and put it in an appropriate folder (perhaps `${HOME}/Exercises/Lecture07 ?!`)

head -30 cytoxseqs.fasta

```
>YP_004300493.1 cytochrome c oxidase subunit I (mitochondrion) [Pristionchus pacificus]
MNMFKKYQGGLAVWLESSNHKDITLYFIFGLWSGMVGTSLSLIIRLELAKPGFFLGNGQLYNSVLTAAH
ILMIFFMVMPMIGGFGNWLLPLMLGAPDMSFPRNNLSFWLLPTAMFLILDSSFVDMGCCGTSWTVYPPL
STMGHPGSSVDLAIFSLHCAGLSSILGGINFMCTTKNLRSSSISLEHMSLFVWTVFVTVFLLVLSLPVLA
GAITMLLTDRNLNTSFFDPSTGGNPLIYQHLFWFFGHPEVYIILILPAFGIISQSTLYLTGKKKEVFGSLGM
VYAILSIGLIGCVVWAHHMYTVGMDLDSRAYFTAATMVIAPTGVKVFVSWLATLFGMKMVFQPVLLWVLG
FIFLFTVGGLTGVLNSSLDIILHDTYYVVSHFHYVLSLGAVFGIFTGISLWWSFITGFVYDKMMMSVV
FILMFIVGNLTFFPLHFAGLHGFPRKYVDYPDVYSIWNIVSSYGSIMISTFALFLFIYVLMESFFSYRLVL
```

NDNEFVNSSPEYSMSGYVFGHSYQSEIYFSSSSSLKH

```
>ADZ52296.1 cytochrome c oxidase subunit I (mitochondrion) [Pristionchus pacificus]
MNMFKKYQGGLAVWLESSNHKDIGTLYFIFGLWSGMVGTSLSLIIRLELAKPGFFLGNGQLYNSVLTAAH
ILMIFFMVMPMIGGFGNWLLPLMLGAPDMSFPRLNNLSFWLLPTAMFLILDSSFVDMGCGTSWTVYPPL
STMGHPPGSSVDLAIFSLHCAGLSSILGGINFMCTTKNLRSSSISLEHMSLFVWTVFVTVFLLVLSLPVLA
GAITMLLTDRNLNTSFFDPSTGGNPLIYQHLFWFFGHPEVYILILPAFGIISQSTLYLTGKKEVFGSLGM
VYAILSIGLIGCVVAHHMYTVGMDLDSRAYFTAATMVIAPVTGVKVFSLATLFGMKMVFPVLLWVLG
FIFLFTVGGLTGVLNSSLDIILHDTYYVVSFHVYVLSLGAVFGIFTGISLWWSFITGFVYDKMMMSVV
FILMFIGVNLTFPPLHFAGLHGFPRKYVDYDPVYSIWNIVSSYGSIMISTFALEFLFIYVLMEFFSYRLVL
NDNEFVNSSPEYSMSGYVFGHSYQSEIYFSSSSSLKH
```

```
>XVF97250.1 cytochrome c oxidase subunit I (mitochondrion) [Travassosius rufus]
MNMFKKYQGGLSVWLESSNHKDIGTLYLLFGLWSGMVGTSLSLIIRLELAKPGVLLGNGQLYNSIITAAH
ILMIFFMVMPMIGGFGNWMVPLMLGAPDMSFPRLNNLSFWLLPTAMFLILDSCFVDMGCGTSWTVYPPL
STLGHPPGSSVDLAIFSLHCAGLSSILGGINFMCTTKNLRSSSISLEHMSLFVWTVFVTVFLLVLTLPVLA
GAITMLLTDRNLNTSFFDPSTGGNPLIYQHLFWFFGHPEVYILILPAFGIISQSTLYLTGKKEVFGSLGM
VYAILSIGLIGCVVAHHMFTVGMDLDSRAYFTAATMVIAPVTGVKVFSLATLFGMKMVLQPLLLWVLG
FIFLFTIGGLTGVLNSSLDIILHDTYYVVSFHVYVLSLGAVFGIFTGISLWWSFMTGFVYDKLMMMAV
FFLMFFGVNLTFPPLHFAGLHGFPRKYMDYDPVYSVWNVMSSYGSIMVSVFAMFMFLFVLVESFFSYRLVL
VDNEFVNSSPECSYSSYVFGHSYQSDIYFSSTYLKD
```

How many sequences do you have in the file?!

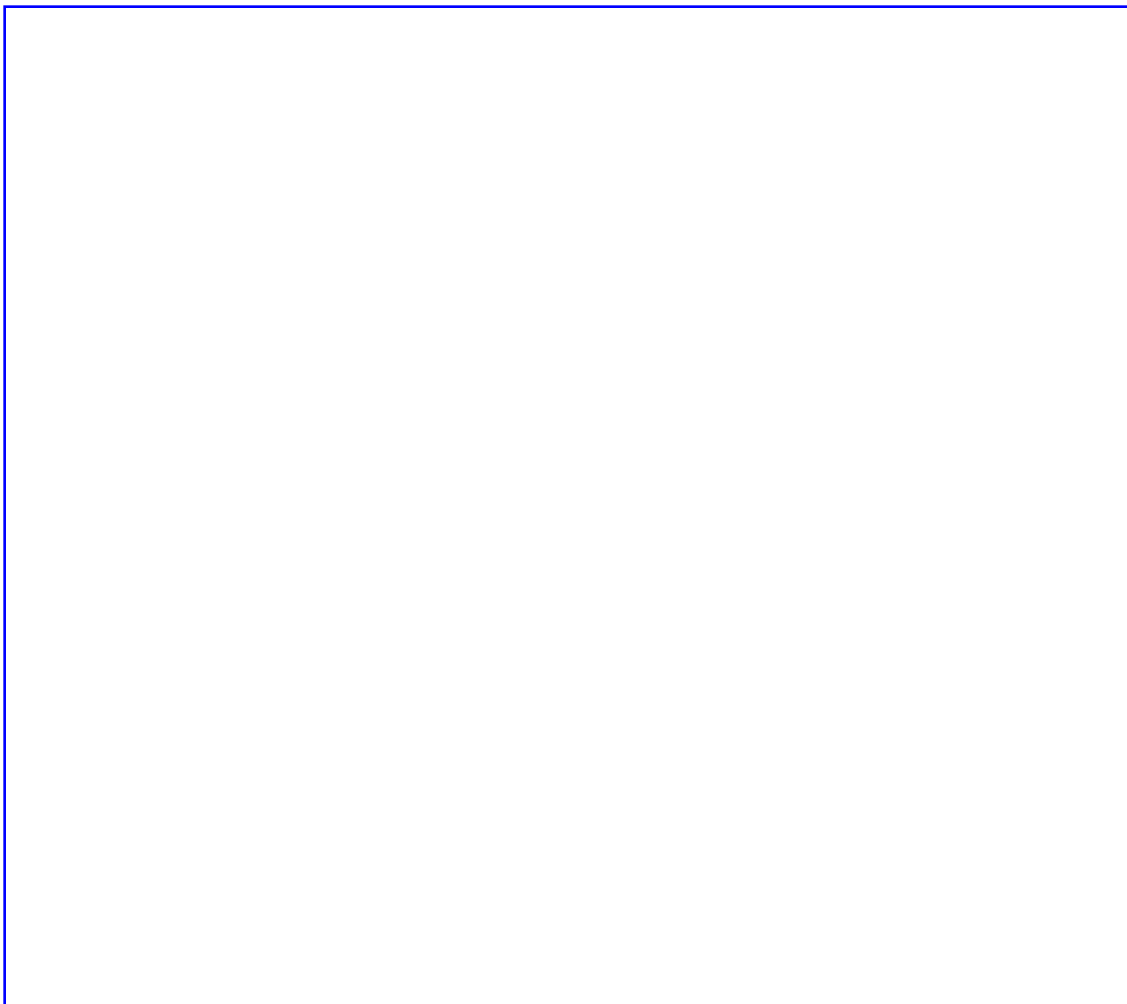
grep -c ">" cytoxseqs.fasta

827

🏠 Alignment of the sequences: EBI Clustal alignment server <http://www.ebi.ac.uk/Tools/msa/clustalo/>

Clustering alignments will help to identify which, if any, regions of the downloaded proteins are similar in sequence/characteristics. There are many tools available to do this, but **clustalo** is probably one of the better ones.

All we need to do is click on Clustalo, upload our sequences, choose parameters, put our email address in, and set it running...



The outputs from Clustal Omega for the run I did recently

These outputs are stored by EBI for 7 days only (but you could of course do your own...)!

[clustalo-I20251002-153624-0107-79514553-p1m](#)

The screenshot displays the Clustal Omega web interface. At the top, there is a navigation bar with links for Research, Training, About us, Search, EMBL-EBI home, and Services. The main heading reads "Clustal Omega Multiple Sequence Alignment (MSA)". Below this, there are links for Job Dispatcher, Help & Privacy, Your Jobs, Input form, and a Feedback button. A welcome message states: "Welcome to the Job Dispatcher website! If you need assistance or have feedback, please [contact](#) us." The section "Results for Job ID" shows the job ID "clustalo-I20251002-153624-0107-79514553-p1m". To the right of the job ID are buttons for "Copy" and "Resubmission". At the bottom, a cookie banner states: "This website requires cookies, and the limited processing of your personal data in order to function. By using the site you are agreeing to this as outlined in our [Privacy Notice](#) and [Terms of Use](#)." with a link to "I agree, dismiss this banner".

Thinking like a ...

Biologist

How can I download some DNA sequences?

For a particular taxonomic group?

For a particular gene?

How can I align them?

How can I find regions of high/low conservation?

How can I figure out what these regions might do?

Bioinformatician

How can I automate this for any gene?

For any taxon?

How can I automatically repeat the analysis when new sequences are released?

How can I make it faster? more reliable?

Can I write code to enable other people to run this analysis on their data?

The alignment is plain text...

🏠 **EMBOSS**
<http://emboss.sourceforge.net/>

European Molecular Biology Open Software Suite

- Set of small, single-purpose bioinformatics tools
- Can read/write many common formats
- Local or online
- **all fully installed on bioinfmsc7 and bioinfmsc8**

EMBOSS Explorer: GUI version

<https://www.bioinformatics.nl/cgi-bin/emboss/>

[\[sort alphabetically \]](#)

ALIGNMENT
[extractalign](#)

ALIGNMENT
CONSENSUS
[cons](#)
[consambig](#)
[megamerger](#)
[merger](#)

ALIGNMENT
DIFFERENCES
[diffseq](#)

ALIGNMENT
DOT PLOTS
[dotmatcher](#)
[dotpath](#)
[dottup](#)
[polydot](#)

ALIGNMENT
GLOBAL
[est2genome](#)
[needle](#)
[needleall](#)
[stretcher](#)

ALIGNMENT
LOCAL
[matcher](#)
[seqmatchall](#)
[supermatcher](#)
[water](#)
[wordfinder](#)

EMBOSS explorer

Welcome to EMBOSS explorer, a graphical user interface to the [EMBOSS](#) suite of bioinformatics tools.

To continue, select an application from the menu to the left. Move the mouse pointer over the name of an application in the menu to display a short description. To search for a particular application, use [wosname](#).

For more information about EMBOSS explorer, including how to download and install it locally, visit the [EMBOSS explorer](#) website.

Development of EMBOSS explorer has been supported by the [National Research Council of Canada](#) and [Genome Prairie](#).

EMBOSS Explorer: plotcon

<https://www.bioinformatics.nl/cgi-bin/emboss/plotcon>

EMBOSS explorer

plotcon

Plot conservation of a sequence alignment ([read the manual](#))

Unshaded fields are optional and can safely be ignored. ([hide optional fields](#))

Input section

Select an input sequence. Use one of the following three fields:

1. To access a sequence from a database, enter the USA here:
2. To upload a sequence from your local computer, select it here: No file chosen
3. To enter the sequence data manually, type here:

Comparison matrix file. Use one of the following two fields:

1. To access a standard EMBOSS data file, enter the name here:
(default is *EBLOSUM62* for protein, *EDNAFULL* for nucleic)
2. To upload a data file from your local computer, select it here: No file chosen

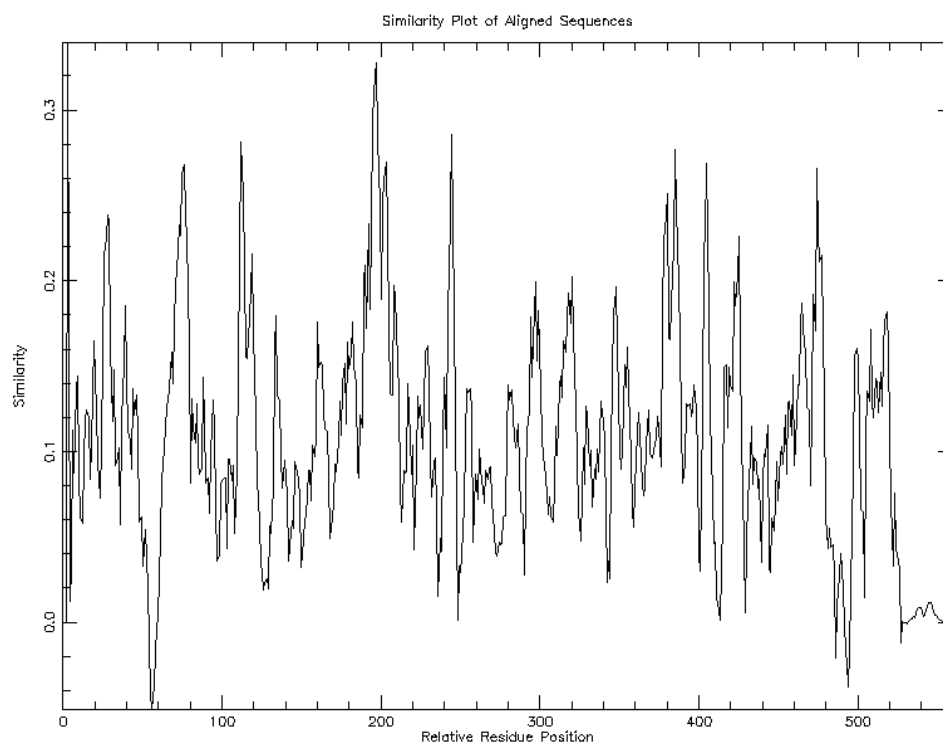
Required section

Window size

Output section

Output graphic format

EMBOSS Explorer: plotcon result



📌EMBOSS plotcon: we don't need a web browser...

EMBOSS is installed on our server!

plotcon -help -verbose

Plot conservation of a sequence alignment
Version: EMBOSS:6.6.0.0

Standard (Mandatory) qualifiers:

<code>[-sequences]</code>	<code>seqset</code>	File containing a sequence alignment
<code>-winsize</code>	<code>integer</code>	[4] Number of columns to average alignment quality over. The larger this value is, the smoother the plot will be. (Any integer value)
<code>-graph</code>	<code>xygraph</code>	[\$EMBOSS_GRAPHICS value, or x11] Graph type (ps, hpgl, hp7470, hp7580, meta, cps, x11, tek, teks, none, data, xterm, svg)

Additional (Optional) qualifiers:

<code>-scorefile</code>	<code>matrix</code>	[EBLOSUM62 for protein, EDNAFULL for DNA] This is the scoring matrix file used when comparing sequences. By default it is the file 'EBLOSUM62' (for proteins) or the file 'EDNAFULL' (for nucleic sequences). These files are found in the 'data' directory of the EMBOSS installation.
-------------------------	---------------------	--

Advanced (Unprompted) qualifiers: (none)

Associated qualifiers:

"-sequences" associated qualifiers

<code>-sbegin1</code>	<code>integer</code>	Start of each sequence to be used
<code>-send1</code>	<code>integer</code>	End of each sequence to be used
<code>-sreverse1</code>	<code>boolean</code>	Reverse (if DNA)
<code>-sask1</code>	<code>boolean</code>	Ask for begin/end/reverse
<code>-snucleotidel</code>	<code>boolean</code>	Sequence is nucleotide
<code>-sprotein1</code>	<code>boolean</code>	Sequence is protein
<code>-slower1</code>	<code>boolean</code>	Make lower case
<code>-supper1</code>	<code>boolean</code>	Make upper case
<code>-scircular1</code>	<code>boolean</code>	Sequence is circular
<code>-squick1</code>	<code>boolean</code>	Read id and sequence only
<code>-sformat1</code>	<code>string</code>	Input sequence format
<code>-iqueryl</code>	<code>string</code>	Input query fields or ID list
<code>-ioffset1</code>	<code>integer</code>	Input start position offset
<code>-sdbname1</code>	<code>string</code>	Database name
<code>-sid1</code>	<code>string</code>	Entryname
<code>-ufo1</code>	<code>string</code>	UFO features
<code>-fformat1</code>	<code>string</code>	Features format
<code>-fopenfile1</code>	<code>string</code>	Features file name

"-graph" associated qualifiers

<code>-gprompt</code>	<code>boolean</code>	Graph prompting
<code>-gdesc</code>	<code>string</code>	Graph description
<code>-gtitle</code>	<code>string</code>	Graph title
<code>-gsubtitle</code>	<code>string</code>	Graph subtitle
<code>-gxtitle</code>	<code>string</code>	Graph x axis title
<code>-gytitle</code>	<code>string</code>	Graph y axis title
<code>-goutfile</code>	<code>string</code>	Output file for non interactive displays
<code>-gdirectory</code>	<code>string</code>	Output directory

General qualifiers:

<code>-auto</code>	<code>boolean</code>	Turn off prompts
<code>-stdout</code>	<code>boolean</code>	Write first file to standard output
<code>-filter</code>	<code>boolean</code>	Read first file from standard input, write first file to standard output
<code>-options</code>	<code>boolean</code>	Prompt for standard and additional values
<code>-debug</code>	<code>boolean</code>	Write debug output to program.dbg
<code>-verbose</code>	<code>boolean</code>	Report some/full command line options
<code>-help</code>	<code>boolean</code>	Report command line options and exit. More

		information on associated and general qualifiers can be found with <code>-help -verbose</code>
<code>-warning</code>	boolean	Report warnings
<code>-error</code>	boolean	Report errors
<code>-fatal</code>	boolean	Report fatal errors
<code>-die</code>	boolean	Report dying program messages
<code>-version</code>	boolean	Report version number and exit

👆 We don't need a web browser for clustalo either

clustalo --help

Clustal Omega - 1.2.4 (AndreaGiacomo)

If you like Clustal-Omega please cite:

Sievers F, Wilm A, Dineen D, Gibson TJ, Karplus K, Li W, Lopez R, McWilliam H, Remmert M, Söding J, Thompson JD, Higgins DG
Fast, scalable generation of high-quality protein multiple sequence alignments using Clustal Omega.

Mol Syst Biol. 2011 Oct 11;7:539. doi: 10.1038/msb.2011.75. PMID: 21988835.

If you don't like Clustal-Omega, please let us know why (and cite us anyway).

Check <http://www.clustal.org> for more information and updates.

Usage: clustalo [-hv] [-i {<file>,-}] [--hmm-in=<file>]... [--hmm-batch=<file>] [--dealign] [--profile1=<file>] [--profile2=

A typical invocation would be: clustalo -i my-in-seqs.fa -o my-out-seqs.fa -v
See below for a list of all options.

Sequence Input:

-i, --in, --infile=<file>,-) Multiple sequence input file (- for stdin)
--hmm-in=<file> HMM input files
--hmm-batch=<file> specify HMMs for individual sequences
--dealign Dealign input sequences
--profile1, --p1=<file> Pre-aligned multiple sequence file (aligned columns will be kept fix)
--profile2, --p2=<file> Pre-aligned multiple sequence file (aligned columns will be kept fix)
--is-profile disable check if profile, force profile (default no)
-t, --seqtype={Protein, RNA, DNA} Force a sequence type (default: auto)
--infmt={a2m=fa[sta],clu[stal],msf,phy[lip],selex,st[ockholm],vie[nna]} Forced sequence input file format (default: auto)

Clustering:

--distmat-in=<file> Pairwise distance matrix input file (skips distance computation)
--distmat-out=<file> Pairwise distance matrix output file
--guidetree-in=<file> Guide tree input file (skips distance computation and guide-tree clustering step)
--guidetree-out=<file> Guide tree output file
--pileup Sequentially align sequences
--full Use full distance matrix for guide-tree calculation (might be slow; mBed is default)
--full-iter Use full distance matrix for guide-tree calculation during iteration (might be slowish; mBed is
--cluster-size=<n> soft maximum of sequences in sub-clusters
--clustering-out=<file> Clustering output file
--trans=<n> use transitivity (default: 0)
--posterior-out=<file> Posterior probability output file
--use-kimura use Kimura distance correction for aligned sequences (default no)
--percent-id convert distances into percent identities (default no)

Alignment Output:

-o, --out, --outfile={file,-} Multiple sequence alignment output file (default: stdout)
--outfmt={a2m=fa[sta],clu[stal],msf,phy[lip],selex,st[ockholm],vie[nna]} MSA output file format (default: fasta)
--residuenumber, --resno in Clustal format print residue numbers (default no)
--wrap=<n> number of residues before line-wrap in output
--output-order={input-order,tree-order} MSA output order like in input/guide-tree

Iteration:

--iterations, --iter=<n> Number of (combined guide-tree/HMM) iterations
--max-guidetree-iterations=<n> Maximum number of guidetree iterations
--max-hmm-iterations=<n> Maximum number of HMM iterations

Limits (will exit early, if exceeded):

--maxnumseq=<n> Maximum allowed number of sequences
--maxseqlen= Maximum allowed sequence length

Miscellaneous:

--auto Set options automatically (might overwrite some of your options)
--threads=<n> Number of processors to use
--pseudo=<file> Input file for pseudo-count parameters
-l, --log=<file> Log all non-essential output to this file
-h, --help Print this help and exit
-v, --verbose Verbose output (increases if given multiple times)
--version Print version information and exit
--long-version Print long version information and exit
--force Force file overwriting

Thinking like a ...

Biologist and bioinformatician working together

How can I download some DNA sequences?

For a particular taxonomic group?

For a particular gene?

How can I align them?

How can I find regions of high/low conservation?

How can I figure out what these regions might do?

How can I automate this for any gene? I'm getting ideas!

For any taxon? I'm getting ideas!

How can I automatically repeat the analysis when new sequences are released?

How can I make it faster? more reliable? I'm getting ideas!

Can I write code to enable other people to run this analysis on their data? I'm getting ideas!

What have we seen so far today?

Databases

NCBI Taxonomy

GenBank

Formats

GenBank

Fasta

Clustal

Tools

Clustal

EMBOSS

Links to many others

📌 More on NCBI's eutils...

Remember doing this next command as part of the [BLAST lecture](#)? We used it to get a bit out of a bigger sequence to then use as a query for BLAST against our "nem" database.

```
https://eutils.ncbi.nlm.nih.gov/entrez/eutils/efetch.fcgi?  
db=nuccore&id=NC_015119.1&strand=1&seq_start=1426&seq_stop=2962&rettype=fasta&retmode=text
```

As mentioned when we were doing the `wget` retrieval of that sequence:

```
wget -O testsequence.fasta "https://eutils.ncbi.nlm.nih.gov/entrez/eutils/efetch.fcgi?  
db=nuccore&id=NC_015119.1&strand=1&seq_start=1426&seq_stop=2962&rettype=fasta&retmode=te
```

there are many NCBI utilities to make doing this kind of thing much easier. These are described in more detail at, for example:

- <https://www.ncbi.nlm.nih.gov/books/NBK25501/>
- <https://www.ncbi.nlm.nih.gov/books/NBK1058/>

The above `wget` line was great for getting one sequence, but what about getting a larger dataset?

Let's look for `all` the nucleotide sequences for *Cosmoscarta*, using our browser.



[http://eutils.ncbi.nlm.nih.gov/entrez/eutils/esearch.fcgi?
db=nucleotide&term=Cosmoscarta&usehistory=y](http://eutils.ncbi.nlm.nih.gov/entrez/eutils/esearch.fcgi?db=nucleotide&term=Cosmoscarta&usehistory=y)

The `usehistory=y` parameter will generate the Web environment (`&WebEnv`) and query key (`&query_key`) parameters that will specify the location of the retrieved GI numbers (**NOTE!**) on the Entrez history server at NCBI.

The default `XML` output will look something like this (depending a bit on your browser, and when you did the search)

Reformatted to look nice (it's actually only a couple of lines)!

```
<eSearchResult>
<Count>131</Count>
<RetMax>20</RetMax>
<RetStart>0</RetStart>
<QueryKey>1</QueryKey>
<WebEnv>MCID_652278055dd7ba5b3618cb25</WebEnv>
<IdList>
<Id>2500442045</Id>
<Id>2500442021</Id>
<Id>2500442019</Id>
<Id>2500442003</Id>
```

```
<Id>2500441995</Id>
<Id>2500441993</Id>
<Id>2500441957</Id>
<Id>2235955617</Id>
<Id>2235955616</Id>
<Id>2235955615</Id>
<Id>2235955614</Id>
<Id>2235955575</Id>
<Id>2235955574</Id>
<Id>2235955573</Id>
<Id>2235955572</Id>
<Id>2235955571</Id>
<Id>2235955569</Id>
<Id>2231887853</Id>
<Id>2231887852</Id>
<Id>2198914917</Id>
</IdList>
<TranslationSet>
<Translation>
<From>Cosmoscarta</From>
<To>"Cosmoscarta" [Organism] OR Cosmoscarta[All Fields]</To>
</Translation>
</TranslationSet>
<TranslationStack>
<TermSet>
<Term>"Cosmoscarta" [Organism]</Term>
<Field>Organism</Field>
<Count>129</Count>
<Explode>Y</Explode>
</TermSet>
<TermSet>
<Term>Cosmoscarta[All Fields]</Term>
<Field>All Fields</Field>
<Count>131</Count>
<Explode>N</Explode>
</TermSet>
<OP>OR</OP>
<OP>GROUP</OP>
</TranslationStack>
<QueryTranslation>"Cosmoscarta" [Organism] OR Cosmoscarta[All Fields]</QueryTranslation>
</eSearchResult>
```

📌 Search done, so now what??

We have done the "esearch" utility bit, so now we use the output of esearch as input for the "efetch" utility to actually get the sequences back!

To do this, the URL needs the query key number (`&query_key`) and the web environment (`&WebEnv`) string that we generated doing the initial esearch command.

Let's specify the retrieval type `rettype` as "uilist" and return mode `retmode` as "text".

Here is our efetch URL

```
http://eutils.ncbi.nlm.nih.gov/entrez/eutils/efetch.fcgi?  
db=nucleotide&query_key=1&WebEnv=MCID_652278055dd7ba5b3618cb25&rettype=uilist&retmode=text
```

Here is our (my) output

```
2500442045  
2500442021  
2500442019  
2500442003  
2500441995  
2500441993  
2500441957  
2235955617  
2235955616  
2235955615  
...  
300248089  
300248008  
300248007  
300247999  
53828034  
53828000  
12002768
```

We actually wanted fasta format, so we just need to change the `rettype` to fasta

Here is our (my) revised URL (one line)

```
http://eutils.ncbi.nlm.nih.gov/entrez/eutils/efetch.fcgi?  
db=nucleotide&query_key=1&WebEnv=MCID_652278055dd7ba5b3618cb25&rettype=fasta&retmode=text
```

Here is our (my) output (well, part of it...!)

```
>OM721651.1 Cosmoscarta sp. voucher C-00132 histone 3 (H3) mRNA, partial cds  
CCGCAAATCGACCGGAGGGAAAGCGCCTAGGAAGCAGCTGGCTACTAAGGCGGCCAGGAAGAGCGCTCCA  
GCCACCGGCGTAGTGATGAAGCCTCACCGTTACAGGCCCGGTACGGTAGCTCTCCGCNAGTTCTGCCGTT  
ACCAGACGAGCACGGAGCTTCTGATCCGCAAGCTCCCTTTCCAAAGGTTGGTCCGAGAAATCGCGCAAGA  
CTTCAT  
  
>OM721639.1 Cosmoscarta pulchella voucher C-00129 histone 3 (H3) mRNA, partial cds  
AGCGCCGAGGAAGCAGCTGGCAACTAAGGCGGCCAGGAAGAGCGCTCCAGCCACCGGCGGAGTGAAGAAG  
CCTCACCGTTACAGGCCAGGTACGGTGGCTCTCCGCGAGATTCGCCGTTACCAGAAGAGCACCGAGCTTC  
TGATCCGCAAGCTCCCGTTCCAACGGTTGGTCCGTGAGATCGCGCAGGACTTCAAGACGGATCTCCGTT  
CCAGAGCTCGGCGGTGATGGCTTTCAGGAGGCGAGCGAA  
... and lots more lines!
```

That works, so let's "automate" it for use in a shell using `wget` !

We'll write a small shell script to get `query_key` and `WebEnv`

First, define a couple of variables that we know already

`beast="Cosmoscarta"`

`db="nucleotide"`

Run `wget` , get the portions of the XML output we want "on the fly"

```
efetchstuff=$(wget -qO- "http://eutils.ncbi.nlm.nih.gov/entrez/eutils/esearch.fcgi?db=nucleotide&term=$beast&retmax=20&retstart=0" | egrep "QueryKey|WebEnv")
```

`echo ${efetchstuff}`

```
<eSearchResult><Count>131</Count><RetMax>20</RetMax><RetStart>0</RetStart><QueryKey>1</QueryKey><WebEnv>MCID_6522791d874b5d29d530daea
```

We can use IFS to split the line into fields, look what it does!

`IFS="/><"`

`echo ${efetchstuff}`

```
eSearchResult Count 131 Count RetMax 20 RetMax RetStart 0 RetStart QueryKey 1 QueryKey WebEnv
```

Just use `awk` to get the bits we want!

`echo ${efetchstuff} | awk '{print $(NF-5),$(NF-2)}'`

`1 MCID_6522791d874b5d29d530daea`

We can also get the bits we want to make the variables we want

`QueryKey=$(echo ${efetchstuff} | awk '{print $(NF-5)}')`

`echo ${QueryKey}`

`1`

`WebEnv=$(echo ${efetchstuff} | awk '{print $(NF-2)}')`

`echo ${WebEnv}`

`MCID_6522791d874b5d29d530daea`

Reset `${IFS}` to its defaults, while we remember....

`unset IFS`

Now we can do the `efetch` component of the search

```
wget -qO cosmoscarta_sequences.fasta \  
"http://eutils.ncbi.nlm.nih.gov/entrez/eutils/efetch.fcgi?db=${db} &query_key=${QueryKey} &WebEnv=
```

Check what we got back

`head -3 cosmoscarta_sequences.fasta`

```
>OM721651.1 Cosmoscarta sp. voucher C-00132 histone 3 (H3) mRNA, partial cds  
CCGCAAATCGACCGGAGGGAAAGCGCCTAGGAAGCAGCTGGCTACTAAGGCGGCCAGGAAGAGCGCTCCA  
GCCACCGGCGTAGTGATGAAGCCTCACCGTTACAGGCCCGGTACGGTAGCTCTCCGCNAGTTCTGCCGTT
```

How many sequences?

`grep -c ">" cosmoscarta_sequences.fasta`

131

What sequences were they?

Get the first three sequence header lines using grep matches

`grep -m3 ">" cosmoscarta_sequences.fasta`

```
>OM721651.1 Cosmoscarta sp. voucher C-00132 histone 3 (H3) mRNA, partial cds  
>OM721639.1 Cosmoscarta pulchella voucher C-00129 histone 3 (H3) mRNA, partial cds  
>OM721638.1 Cosmoscarta septempunctata voucher C-00126 histone 3 (H3) mRNA, partial cds
```

That works, but it wasn't that much fun, was it? 😞

🏠Entrez Direct: E-utilities on the UNIX Command Line

It's good practice writing code to do things you want, but sometimes it's just easier/quicker to use wheels that have already been invented! 🛞

One of these is Entrez Direct (EDirect), which provides UNIX command line alternative to E-utilities; it was mentioned about half way down the <https://www.ncbi.nlm.nih.gov/books/NBK25501/> web page (from the [BLAST lecture](#))...

EDirect has its own web page: <https://www.ncbi.nlm.nih.gov/books/NBK179288/>

EDirect will run on UNIX and Macintosh computers that have the Perl language installed, and under the Cygwin UNIX-emulation environment on Windows PCs.

The EDirect www page tells you how you can install it yourself on the MSc server.

The link to all things about edirect is

<https://www.ncbi.nlm.nih.gov/books/NBK179288/>

PLEASE DO THIS!

Install line, from wherever you are, it doesn't matter...

```
sh -c "$(wget -q ftp://ftp.ncbi.nlm.nih.gov/entrez/entrezdirect/install-edirect.sh -O -)"
```

Entrez Direct has been successfully downloaded and installed.

It will also ask the following:

Entrez Direct has been successfully downloaded and installed.

In order to complete the configuration process, please execute the following:

```
echo "source ~/.bash_profile" >> ${HOME}/.bashrc
```

or manually edit the PATH variable assignment in your .bash_profile file.

Would you like to do that automatically now? [y/N]

y

OK, done.

What would that do?!

Will it work in an interactive shell and/or a login shell?!

```
cat ~/.bash_profile | grep edirect
```

...what did you get?!...

```
cat ~/.bashrc | grep edirect
```

...what did you get?!...

Check the esearch programme

esearch -h

esearch 24.7

Query Specification

-db	Database name
-query	Query string

Spell Check

-spell	Correct misspellings in query
--------	-------------------------------

Query Translation

-translate	Show automatic term mapping
-component	Individual term mapping items

Document Order

-sort	Result presentation order
-------	---------------------------

Sort Choices by Database

gene	Chromosome, Gene Weight, Name, Relevance
geoprofiles	Default Order, Deviation, Mean Value, Outliers, Subgroup Effect
pubmed	First Author, Journal, Last Author, Pub Date, Recently Added, Relevance, Title
(sequences)	Accession, Date Modified, Date Released, Default Order, Organism Name, Taxonomy ID
snp	Chromosome Base Position, Default Order, Heterozygosity, Organism, SNP_ID, Success Rate

Note

All efilter shortcuts can also be used with esearch

Examples

```
esearch -db pubmed -query "opsin gene conversion OR tetrachromacy"

esearch -db pubmed -query "Garber ED [AUTH] AND PNAS [JOUR]"

esearch -db nuccore -query "MatK [GENE] AND NC_0:NC_999999999 [PACC]"

esearch -db protein -query "amyloid* [PROT]" |
elink -target pubmed -label prot_cit |
esearch -db gene -query "apo* [GENE]" |
elink -target pubmed -label gene_cit |
esearch -query "(#prot_cit) AND (#gene_cit)" |
efetch -format docsum |
xtract -pattern DocumentSummary -element Id Title
```

Here are some examples for how we might get all "organism is *Cosmoscarta*" records in the NCBI nucleotide database using `esearch`

```
esearch -db nucleotide -query "Cosmoscarta[organism]"
```

```
<ENTREZ_DIRECT>
  <Db>nucleotide</Db>
  <WebEnv>MCID_65227daa9ff8291591798109</WebEnv>
  <QueryKey>1</QueryKey>
  <Count>129</Count>
  <Step>1</Step>
  <Elapsed>0</Elapsed>
</ENTREZ_DIRECT>
```

```
# Use esearch then efetch to get the UID values
```

```
esearch -db nucleotide -query "Cosmoscarta[organism]" | efetch -format uid >
Cosmoscarta.nuc.gis
```

```
head -n5 Cosmoscarta.nuc.gis
```

```
12002768
53828000
53828034
53828082
300247880
```

```
# Use esearch then efetch to get the accession values
```

```
esearch -db nucleotide -query "Cosmoscarta[organism]" | efetch -format acc >
Cosmoscarta.nuc.acc
```

```
head Cosmoscarta.nuc.acc
```

```
OM721639.1
OM721638.1
OM721630.1
OM721626.1
OM721625.1
OM721607.1
OM721651.1
ON455170.1
ON455169.1
ON455168.1
```

```
wc -l Cosmoscarta.nuc.acc
```

```
129 Cosmoscarta.nuc.acc
```

```
# Use esearch then efetch to get the fasta seqs
```

```
esearch -db nucleotide -query "Cosmoscarta[organism]" | efetch -format fasta >
Cosmoscarta.nuc.fa
```

```
head -n5 Cosmoscarta.nuc.fa
```

```
>>OM721639.1 Cosmoscarta pulchella voucher C-00129 histone 3 (H3) mRNA, partial cds
AGCGCCGAGGAAGCAGCTGGCAACTAAGGCGGCCAGGAAGAGCGCTCCAGCCACCGGCGGAGTGAAGAAG
CCTCACCGTTACAGGCCAGGTACGGTGGCTCTCCGCGAGATTGCGCGTTACCAGAAGAGCACCGAGCTTC
TGATCCGCAAGCTCCCGTTCCAACGGTTGGTCCGTGAGATCGCGCAGGACTTCAAGACGGATCTCCGGTT
CCAGAGCTCGGCGGTGATGGCTTTGCAGGAGGCGAGCGAA
```

```
# How many fasta seqs
```

```
grep -c ">" Cosmoscarta.nuc.fa
```

```
129
```

Much easier, huh?!



📌 git and GitHub time!

Status?

git st

On branch master
Your branch is up-to-date with 'Lecture06/master'.

Changes not staged for commit:

(use "git add/rm <file>..." to update what will be committed)
(use "git restore <file>..." to discard changes in working directory)
deleted: ../Lecture04/CodeFiles.tar.gz
modified: ../Lecture06/run_this_script_v1

Untracked files:

(use "git add <file>..." to include in what will be committed)
../BN.out
../BN.sh
../Lecture04/Als_ICA/
../Lecture04/all_the_things_I_did
../Lecture04/outs/
../Lecture05/Country.Mozambique.details
./

no changes added to commit (use "git add" and/or "git commit -a")

What files do we have in the folder?

ls -l

```
Cosmoscarta.nuc.acc  
Cosmoscarta.nuc.fa  
Cosmoscarta.nuc.gis  
cosmoscarta_sequences.fasta  
cytoxseqs.fasta  
myfileofaccessions.txt
```

Add and commit all the files to our repo?

No, this time, let's choose some...

git add -i

```
          staged      unstaged path  
1:    unchanged      +0/-0 Lecture06/run_this_script_v1  
  
*** Commands ***  
1: s tatus      2: u pdate      3: r evert      4: a dd untracked  
5: p atch       6: d iff        7: q uit       8: h elp  
What now> a  
          staged      unstaged path  
1: BN.out  
2: BN.sh  
3: Lecture04/Als_ICA/
```

```
4: Lecture04/all_the_things_I_did
5: Lecture04/outs/someotherfile.sh
6: Lecture05/Country.Mozambique.details
7: Lecture07/Cosmoscarta.nuc.acc
8: Lecture07/Cosmoscarta.nuc.fa
9: Lecture07/Cosmoscarta.nuc.gis
10: Lecture07/cosmoscarta_sequences.fasta
11: Lecture07/cytoseqs.fasta
12: Lecture07/myfileofaccessions.txt
Add untracked>> 10
      staged      unstaged path
1: BN.out
2: BN.sh
3: Lecture04/Als_ICA/
4: Lecture04/all_the_things_I_did
5: Lecture04/outs/someotherfile.sh
6: Lecture05/Country.Mozambique.details
7: Lecture07/Cosmoscarta.nuc.acc
8: Lecture07/Cosmoscarta.nuc.fa
9: Lecture07/Cosmoscarta.nuc.gis
*10: Lecture07/cosmoscarta_sequences.fasta
11: Lecture07/cytoseqs.fasta
12: Lecture07/myfileofaccessions.txt
Add untracked>> 11
      staged      unstaged path
1: BN.out
2: BN.sh
3: Lecture04/Als_ICA/
4: Lecture04/all_the_things_I_did
5: Lecture04/outs/someotherfile.sh
6: Lecture05/Country.Mozambique.details
7: Lecture07/Cosmoscarta.nuc.acc
8: Lecture07/Cosmoscarta.nuc.fa
9: Lecture07/Cosmoscarta.nuc.gis
*10: Lecture07/cosmoscarta_sequences.fasta
*11: Lecture07/cytoseqs.fasta
12: Lecture07/myfileofaccessions.txt
Add untracked>> 7
      staged      unstaged path
1: BN.out
2: BN.sh
3: Lecture04/Als_ICA/
4: Lecture04/all_the_things_I_did
5: Lecture04/outs/someotherfile.sh
6: Lecture05/Country.Mozambique.details
* 7: Lecture07/Cosmoscarta.nuc.acc
8: Lecture07/Cosmoscarta.nuc.fa
9: Lecture07/Cosmoscarta.nuc.gis
*10: Lecture07/cosmoscarta_sequences.fasta
*11: Lecture07/cytoseqs.fasta
12: Lecture07/myfileofaccessions.txt
Add untracked>> s
Huh (s)?
      staged      unstaged path
1: BN.out
2: BN.sh
3: Lecture04/Als_ICA/
4: Lecture04/all_the_things_I_did
5: Lecture04/outs/someotherfile.sh
6: Lecture05/Country.Mozambique.details
* 7: Lecture07/Cosmoscarta.nuc.acc
8: Lecture07/Cosmoscarta.nuc.fa
9: Lecture07/Cosmoscarta.nuc.gis
*10: Lecture07/cosmoscarta_sequences.fasta
*11: Lecture07/cytoseqs.fasta
12: Lecture07/myfileofaccessions.txt
Add untracked>> u
Huh (u)?
      staged      unstaged path
```

```
1: BN.out
2: BN.sh
3: Lecture04/Als_ICA/
4: Lecture04/all_the_things_I_did
5: Lecture04/outs/someotherfile.sh
6: Lecture05/Country.Mozambique.details
* 7: Lecture07/Cosmoscarta.nuc.acc
8: Lecture07/Cosmoscarta.nuc.fa
9: Lecture07/Cosmoscarta.nuc.gis
*10: Lecture07/cosmoscarta_sequences.fasta
*11: Lecture07/cytoseqs.fasta
12: Lecture07/myfileofaccessions.txt
Add untracked>> q
Huh (q)?
      staged      unstaged path
1: BN.out
2: BN.sh
3: Lecture04/Als_ICA/
4: Lecture04/all_the_things_I_did
5: Lecture04/outs/someotherfile.sh
6: Lecture05/Country.Mozambique.details
* 7: Lecture07/Cosmoscarta.nuc.acc
8: Lecture07/Cosmoscarta.nuc.fa
9: Lecture07/Cosmoscarta.nuc.gis
*10: Lecture07/cosmoscarta_sequences.fasta
*11: Lecture07/cytoseqs.fasta
12: Lecture07/myfileofaccessions.txt
Add untracked>> h
Huh (h)?
      staged      unstaged path
1: BN.out
2: BN.sh
3: Lecture04/Als_ICA/
4: Lecture04/all_the_things_I_did
5: Lecture04/outs/someotherfile.sh
6: Lecture05/Country.Mozambique.details
* 7: Lecture07/Cosmoscarta.nuc.acc
8: Lecture07/Cosmoscarta.nuc.fa
9: Lecture07/Cosmoscarta.nuc.gis
*10: Lecture07/cosmoscarta_sequences.fasta
*11: Lecture07/cytoseqs.fasta
12: Lecture07/myfileofaccessions.txt
Add untracked>>
added 3 paths

*** Commands ***
1: status      2: update      3: revert      4: add untracked
5: patch       6: diff        7: quit        8: help
What now> q
Bye

What now> q
Bye.
```

Check the status

git st

On branch master
Your branch is up-to-date with 'Lecture06/master'.

Changes to be committed:

```
(use "git restore --staged <file>..." to unstage)
new file:   Cosmoscarta.nuc.acc
new file:   cosmoscarta_sequences.fasta
new file:   cytoseqs.fasta
```

Changes not staged for commit:

```
(use "git add/rm <file>..." to update what will be committed)
(use "git restore <file>..." to discard changes in working directory)
deleted:    ../Lecture04/CodeFiles.tar.gz
modified:   ../Lecture06/run_this_script_v1
```

Untracked files:

```
(use "git add <file>..." to include in what will be committed)
../BN.out
../BN.sh
../Lecture04/Als_ICA/
../Lecture04/all_the_things_I_did
../Lecture04/outs/
../Lecture05/Country.Mozambique.details
Cosmoscarta.nuc.fa
Cosmoscarta.nuc.gis
myfileofaccessions.txt
```

Commit with a message

git commit -m "edirect seqs related stuff added"

```
[master eae65c6] edirect seqs related stuff added
3 files changed, 12900 insertions(+)
create mode 100644 Lecture07/Cosmoscarta.nuc.acc
create mode 100644 Lecture07/cosmoscarta_sequences.fasta
create mode 100644 Lecture07/cytosxseqs.fasta
```

GitHub: create a repository in your personal account on GitHub

You will also need your personal account GitHub token again (NOT password...)!

Doing it the variable way... use your own details... assuming you have defined them first!

set | grep "myper"

```
mypersonalGitHub_username="B123456-2121"
```

```
mypersonalgittoken="ghp_mKYRL....."
```

```
curl -u ${mypersonalGitHub_username}:${mypersonalgittoken} \
https://api.github.com/user/repos -d '{"name":"BPSM_Lecture07"}'
```

```
{
  "id": 548550084,
  "node_id": "R_kgDOILI1xA",
  "name": "BPSM_Lecture07",
  "full_name": "B123456-2121/BPSM_Lecture07",
  "private": false,
  ...
```

If you run the same command again, you will get something like this as output...:

```
{
  "message": "Repository creation failed.",
  "errors": [
    {
      "resource": "Repository",
      "code": "custom",
      "field": "name",
```



```
    "message": "name already exists on this account"
  }
],
"documentation_url": "https://docs.github.com/rest/repos/repos#create-a-repository-for-t",
"status": "422"
}
```

**# Tell git about our repository on GitHub, and that
we are calling it "Lecture07" (or something!)**

git remote add Lecture07

[https://github.com/\\${mypersonalGitHub_username}/BPSM_Lecture07.git](https://github.com/${mypersonalGitHub_username}/BPSM_Lecture07.git)

What links to GitHub have we now?

git remote

```
Lecture06
Lecture07
origin
```

Push our repository (called master) off to our repository on GitHub

git push -u Lecture07 master

```
Username for 'https://github.com': B123456-2121
Password for 'https://B123456-2121@github.com':
To https://github.com/B123456-2121/BPSM_Lecture07.git
Enumerating objects: 102, done.
Counting objects: 100% (102/102), done.
Delta compression using up to 256 threads
Compressing objects: 100% (95/95), done.
Writing objects: 100% (102/102), 26.57 MiB | 4.80 MiB/s, done.
Total 102 (delta 31), reused 0 (delta 0)
remote: Resolving deltas: 100% (31/31), done.
To https://github.com/B123456-2121/BPSM_Lecture07.git
 * [new branch]      master -> master
Branch 'master' set up to track remote branch 'master' from 'Lecture07'.
```

OR.....

```
...
! [rejected]      master -> master (fetch first)
error: failed to push some refs to 'https://github.com/B123456-2121/BPSM_Lecture07.git'
hint: Updates were rejected because the remote contains work that you do not
hint: have locally. This is usually caused by another repository pushing to
hint: the same ref. If you want to integrate the remote changes, use
hint: 'git pull' before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.
```

Oooooops, actually forgot we had pushed to Lecture06 last time...

git push -u Lecture06 master

```
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 256 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (6/6), 131.10 KiB | 2.11 MiB/s, done.
Total 6 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/B123456-2121/BPSM_Lecture06.git
 eea4f5e..eae65c6  master -> master
branch 'master' set up to track 'Lecture06/master'.
```

Then we changed our minds yet again.... what a mess.... 🙄 🙄 🙄

We want:

- Lecture06 commit to the Lecture06 link
- Lecture07 commit to the Lecture07 link

Easiest (probably) to just remove the two GitHub repos, and remake?

DELETE STUFF

```
curl -u ${mypersonalGitHub_username}:${mypersonalgittoken} -X DELETE  
https://api.github.com/repos/${mypersonalGitHub_username}/BPSM_Lecture06
```

```
curl -u ${mypersonalGitHub_username}:${mypersonalgittoken} -X DELETE  
https://api.github.com/repos/${mypersonalGitHub_username}/BPSM_Lecture07
```

```
git remote rm Lecture06
```

```
git remote rm Lecture07
```

You might have got this?

"Deleting a repository requires admin access. If OAuth is used, the delete_repo scope is required."



github.com/settings/tokens/713391097

GitHub Apps
OAuth Apps
Personal access tokens

Edit personal access token

If you've lost or forgotten this token, you can regenerate it, but be aware that any scripts or applications using this token will need to be updated.

Regenerate token

Note

123456-2121

What's this token for?

Expiration

This token has no expiration date. To set a new expiration date, you must [regenerate the token](#).

Select scopes

Scopes define the access for personal tokens. [Read more about OAuth scopes](#).

☒ repo	Full control of private repositories
☒ repos:status	Access commit status
☒ repos:deployment	Access deployment status
☒ repos:public_repo	Access public repositories
☒ repos:invite	Access repository invitations
☒ security_events	Read and write security events
☐ workflow	Update GitHub Action workflows
☐ write:packages	Upload packages to GitHub Package Registry
☐ read:packages	Download packages from GitHub Package Registry
☐ delete:packages	Delete packages from GitHub Package Registry
☐ admin:org	Full control of orgs and teams, read and write org projects
☐ write:org	Read and write org and team membership, read and write org projects
☐ read:org	Read org and team membership, read org projects

github.com/settings/tokens/713391097

☐ notifications	Access notifications
☐ user	Update ALL user data
☐ read:user	Read ALL user profile data
☐ user:email	Access user email addresses (read-only)
☐ user:follow	Follow and unfollow users
☒ delete_repo	Delete repositories
☐ write:discussion	Read and write team discussions
☐ read:discussion	Read team discussions
☐ admin:enterprise	Full control of enterprises
☐ manage:runners:enterprise	Manage enterprise runners and runner groups
☐ manage:billing:enterprise	Read and write enterprise billing data
☐ read:enterprise	Read enterprise profile data
☐ project	Full control of projects
☐ read:project	Read access of projects
☐ admin:gpg_key	Full control of public user GPG keys
☐ write:gpg_key	Write public user GPG keys
☐ read:gpg_key	Read public user GPG keys
☐ admin:ssh_signing_key	Full control of public user SSH signing keys
☐ write:ssh_signing_key	Write public user SSH signing keys
☐ read:ssh_signing_key	Read public user SSH signing keys

Update token Cancel

REMAKE

```
curl -u ${mypersonalGitHub_username}:${mypersonalgittoken} https://api.github.com/user/repos -d '{"name":"BPSM_Lecture06"}
```

```
curl -u ${mypersonalGitHub_username}:${mypersonalgittoken} https://api.github.com/user/repos -d '{"name":"BPSM_Lecture07"}
```

```
git remote add Lecture06 https://github.com/${mypersonalGitHub_username}/BPSM_Lecture06.git
```

```
git remote add Lecture07 https://github.com/${mypersonalGitHub_username}/BPSM_Lecture07.git
```

```
git remote get-url Lecture06
```

https://github.com/B123456-2121/BPSM_Lecture06.git

git remote get-url Lecture07

https://github.com/B123456-2121/BPSM_Lecture07.git

git log | grep -m2 "commit"

```
commit eae65c6dd66cf180ab318710cc3c2097130aad30
commit eea4f5e119501a6dfa74f1428814ce3b1bd97c58
```

git checkout -b Lec_6_branch

eea4f5e119501a6dfa74f1428814ce3b1bd97c58

```
D      Lecture04/CodeFiles.tar.gz
M      Lecture06/run_this_script_v1
Switched to a new branch 'Lec_6_branch'
```

git log | head -n5

```
eea4f5e Kept bash script output files from Lecture 06
f5e5cc0 Added run_this_script_v1 from BPSM Lecture06
ce2c15a nematode BLASTDB stuff added
19cdd2d Added the example_people_data.tsv file from Lecture03
137c3db Decided to keep blastoutput2.out
```

git push Lecture06 Lec_6_branch

```
Enumerating objects: 110, done.
Counting objects: 100% (110/110), done.
Delta compression using up to 256 threads
Compressing objects: 100% (103/103), done.
Writing objects: 100% (110/110), 26.47 MiB | 9.32 MiB/s, done.
Total 110 (delta 35), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (35/35), done.
To https://github.com/B123456-2121/BPSM_Lecture06.git
 * [new branch]      Lec_6_branch -> Lec_6_branch
```

git checkout master

```
D      Lecture04/CodeFiles.tar.gz
M      Lecture06/run_this_script_v1
Switched to branch 'master'
```

git log | head -n5

```
eae65c6 edirect seqs related stuff added
eea4f5e Kept bash script output files from Lecture 06
f5e5cc0 Added run_this_script_v1 from BPSM Lecture06
ce2c15a nematode BLASTDB stuff added
19cdd2d Added the example_people_data.tsv file from Lecture03
```

git push Lecture07 master

```
Enumerating objects: 116, done.
Counting objects: 100% (116/116), done.
Delta compression using up to 256 threads
Compressing objects: 100% (109/109), done.
Writing objects: 100% (116/116), 26.60 MiB | 9.30 MiB/s, done.
Total 116 (delta 34), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (34/34), done.
```

```
To https://github.com/B123456-2121/BPSM_Lecture07.git
* [new branch]      master -> master
```

Quick checks...

git st

On branch master

Changes not staged for commit:

```
(use "git add/rm <file>..." to update what will be committed)
(use "git restore <file>..." to discard changes in working directory)
    deleted:    ../Lecture04/CodeFiles.tar.gz
    modified:   ../Lecture06/run_this_script_v1
```

Untracked files:

```
(use "git add <file>..." to include in what will be committed)
    ../BN.out
    ../BN.sh
    ../Lecture04/Als_ICA/
    ../Lecture04/all_the_things_I_did
    ../Lecture04/outs/
    ../Lecture05/Country.Mozambique.details
    Cosmoscarta.nuc.fa
    Cosmoscarta.nuc.gis
    myfileofaccessions.txt
```

no changes added to commit (use "git add" and/or "git commit -a")

git branch

```
Lec_6_branch
Version1.2
* master
```

git branch -a

```
Lec_6_branch
Version1.2
* master
remotes/Lecture06/Lec_6_branch
remotes/Lecture07/master
```

git merge Lec_6_branch

Already up-to-date.

git push Lecture07 master

Everything up-to-date

git diff --name-only master Lecture06/Lec_6_branch

```
Lecture07/Cosmoscarta.nuc.acc
Lecture07/cosmoscarta_sequences.fasta
Lecture07/cyttoxseqs.fasta
```

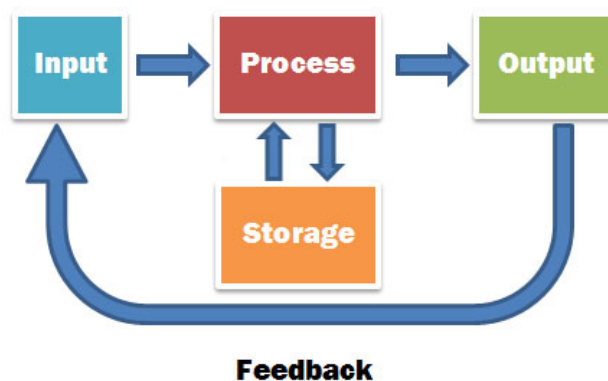

Exercises

No answers to look for today....!

- [Server log in](#)
- [git and GitHub](#)
- [bash scripting](#)
- [Linux commands](#)

🏠.... INSTALL EDIRECT

1. 🏠 Follow the steps in the lecture and make sure that you're able to produce the amino acid conservation map, maybe several times with different window sizes? You'll find that edirect is DEFINITELY the quickest method...!
2. 🏠 Repeat the analysis for a different taxonomic group (primates, maybe?) and gene (Hox-A7 homeobox gene?). Whatever you chose, aim for about 30-600 genes in your selected set, otherwise things might not be sensible, or will take rather longer than you want!



3. 🏠 draw a flowchart (on paper) showing the `plotcon` pipeline in general outline.

- what are the intermediate steps?
- what format(s) do the data have to be in?
- what checks would need to be done during the process to ensure that the workflow is doing what you think it is?

4. Think about how you might automate this pipeline. What would the inputs and outputs need to be for each step? It's just like the optional [MyFirstPipeline](#) ICA!

Have your say...

The "Postbox of Truth": is your chance to give anonymous "in course" feedback/comments. Your comments are completely anonymous and confidential, so say what you think, irrespective of whether the comments are positive or otherwise! You can use this multiple times too: I use it throughout the course to enable you to give context-relevant feedback/make suggestions.

Just click the image!



Sources used